

L8: Midterm 1 review

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What is on Midterm 1?

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Content from L01-L07:

- ▶ PPDAC and types of problems
- ▶ Variable types and unit of analysis
- ▶ Visualizing and summarizing one variable
- ▶ Scatterplots, correlation, and simple linear regression
- ▶ Two categorical variables (tables, conditional distributions, Simpson's paradox)
- ▶ Sampling, observational studies, experiments, bias, ethics
- ▶ Basic dplyr / ggplot2 literacy (recognize and fill blanks)

How to use this review

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- ▶ These worked examples are **practice**, not the exam.
- ▶ Focus on **what skill each question is testing**.
- ▶ For every abstract: name the **problem type**, **design**, **exposure/outcome**, and **variable types**.
- ▶ For every plot/table: describe **shape / center / spread**, or **form / direction / strength / outliers**.

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Problem → Plan → Data → Analysis → Conclusion

In the **Plan**, you decide:

- ▶ census vs sample
- ▶ observational vs experimental
- ▶ timing (cross-sectional vs longitudinal; prospective vs retrospective)

Three types of problems

Type	Goal
Descriptive	Summarize / visualize what is happening
Predictive	Forecast a future or unobserved value
Causative	Estimate the effect of an exposure / intervention

Ask: *Are we describing, forecasting, or claiming “what if we change X”?*

Practice: identify the problem type

- A. A pharmacy wants to stock influenza vaccine for November using last year's weekly shot counts.
- B. A dashboard shows mean PH142 midterm scores by major and year.
- C. Researchers evaluate whether a workplace caffeine-policy change reduces afternoon migraine days.

Try to classify A / B / C before revealing answers on the next slide.

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- ▶ A: Predictive – using past data to project future demand
- ▶ B: Descriptive – summarizing existing scores
- ▶ C: Causative – assessing the effect of a policy change

Another set (different wording)

- D. You map county tree-canopy percent and summer ER heat visits to see the current pattern.
 - E. A clinic builds a model of next week's appointment no-shows from last month's electronic records.
 - F. An RCT assigns neighborhoods to shade canopies vs usual infrastructure and compares heat-stroke admissions.
- D descriptive · E predictive · F causative

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The **unit of analysis** is who/what each row represents.

Dataset idea	Unit
One row per lake with mercury level	lake
One row per clinic visit	visit (not patient, if patients recur)
One row per county-year	county-year

Always check whether repeated measures collapse multiple observations into one row.

Types of variables

- ▶ **Categorical**
 - ▶ **Nominal**: unordered labels (blood type; site A/B/C)
 - ▶ **Ordinal**: ordered categories (cancer stage I-IV; Likert satisfaction)
- ▶ **Quantitative**
 - ▶ **Discrete**: countable integers (number of ED visits)
 - ▶ **Continuous**: measured on a continuum (BMI, wait time in minutes)

Numbers can still be categorical (e.g., stage coded 1-4 = ordinal).

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Practice: classify variables

A dataset on city parks has:

1. park ID
 2. neighborhood name
 3. park size (acres)
 4. playground quality rating: poor / fair / good / excellent
 5. annual visitors
 6. has restroom (yes/no)
- ▶ 1 ID/label · 2 nominal · 3 continuous · 4 ordinal
 - ▶ 5 discrete (or continuous if estimated) · 6 nominal / binary

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Worked abstract (NOT the exam)

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Adapted style of an epi methods abstract (fictional setting):

We examined whether residential proximity to a new light-rail station was associated with change in self-reported weekly physical activity among working-age adults in Metro City. Adults were enrolled in 2018 (before station opening) and followed annually through 2022. Station proximity was classified as < 0.5 mile, 0.5 - 1 mile, or > 1 mile from home. Physical activity minutes were obtained from yearly questionnaires. Analyses adjusted for age, income, and baseline activity.

Check your knowledge

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1. Problem type: descriptive / predictive / causative?
2. Observational or experimental?
3. Cross-sectional or longitudinal?
4. Exposure variable + type?
5. Outcome variable + type?
6. Name one confounder already mentioned; propose one more.

Suggested answers

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1. Causative (association interpreted toward impact of proximity / access)
2. Observational (exposure not assigned by investigators)
3. Longitudinal / prospective (baseline before opening, then follow-up)
4. Proximity category – ordinal
5. Weekly activity minutes – continuous (or discrete if counted in whole minutes)
6. Age, income, baseline activity already adjusted; e.g. car ownership, neighborhood walkability

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When an abstract / methods paragraph appears, mark all that apply:

- ▶ Observational / Experimental
- ▶ Cross-sectional / Longitudinal
- ▶ Prospective / Retrospective
- ▶ Cohort (sampled by exposure) / Case-control (sampled by outcome)
- ▶ Target population / Study population / Sample

These labels are frequently tested – practice justifying each choice from the text.

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Core dplyr verbs

Goal	Function
Keep / drop columns	<code>select()</code>
Keep / drop rows	<code>filter()</code>
Create / change a variable	<code>mutate()</code>
Sort rows	<code>arrange()</code>
Collapse by group	<code>group_by() + summarize()</code>
Count rows	<code>nrow()</code> or <code>count()</code>

Pipe: `data %>% verb(...)`.

Remember to **assign** if you want a permanent change: `data <- data %>% ...`

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Common exam trap: select vs filter

Suppose `visits` has columns `age`, `sex`, `county`, `ed_status`.

Remove the `county` variable:

```
visits <- visits %>% select(-county)
```

Keep only adults $\text{age} \geq 18$:

```
visits <- visits %>% filter(age >= 18)
```

`-column` inside `filter()` is wrong; `filter` chooses rows.

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Quick practice

Fill in the blanks:

```
# create BMI, then keep adults with BMI >= 30  
clinic <- clinic %>%  
  _____(bmi = weight_kg / (height_m^2)) %>%  
  _____(bmi >= 30)
```

Answers: mutate, then filter.

Matching geoms to data

Data type	Typical geom
Counts / proportions of categories	<code>geom_bar()</code>
Distribution of one quantitative variable	<code>geom_histogram()</code> or <code>geom_boxplot()</code>
Relationship of two quantitative variables	<code>geom_point()</code> (+ optional <code>geom_smooth</code> / <code>geom_abline</code>)
Time series (one series)	<code>geom_line()</code>

Comparing groups of a quantitative variable: prefer [side-by-side boxplots](#), not a bar chart of raw numbers unless you intentionally plot summaries.

Recognizing ggplot code

```
ggplot(dat, aes(x = wait_min)) +  
  geom_histogram(binwidth = 10)
```

```
ggplot(dat, aes(x = clinic, y = wait_min)) +  
  geom_boxplot()
```

```
ggplot(dat, aes(x = canopy_pct, y = night_temp)) +  
  geom_point()
```

Ask: *What is on x / y? Continuous or categorical? What relationship is shown?*

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Shape, center, spread

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For a quantitative distribution, always comment on:

1. **Shape**: symmetric, right-skewed, left-skewed; unimodal / multimodal
2. **Center**: mean and / or median
3. **Spread**: IQR, SD / variance, range
4. **Outliers**: unusually large or small values

Mean vs median

- ▶ Symmetric, unimodal $\Rightarrow$ mean $\approx$ median
- ▶ Right-skewed (long right tail, extreme high values) $\Rightarrow$ mean $>$ median
- ▶ Left-skewed $\Rightarrow$ mean $<$ median

Five-number summary: min, Q1, median, Q3, max.

$IQR = Q3 - Q1$.

Practice: infer skew from a five-number summary

Emergency department wait times (minutes):

$\min = 5$, $Q1 = 12$, $\text{median} = 20$, $Q3 = 35$, $\max = 180$, $\text{mean} = 42$

- ▶ Median = 20, mean = 42 > median, max far above Q3
- ▶ Expect a **right-skewed** distribution

Compare: if the five-number summary were 10, 18, 22, 26, 30 with mean ≈ 21 , you would expect approximate **symmetry**.

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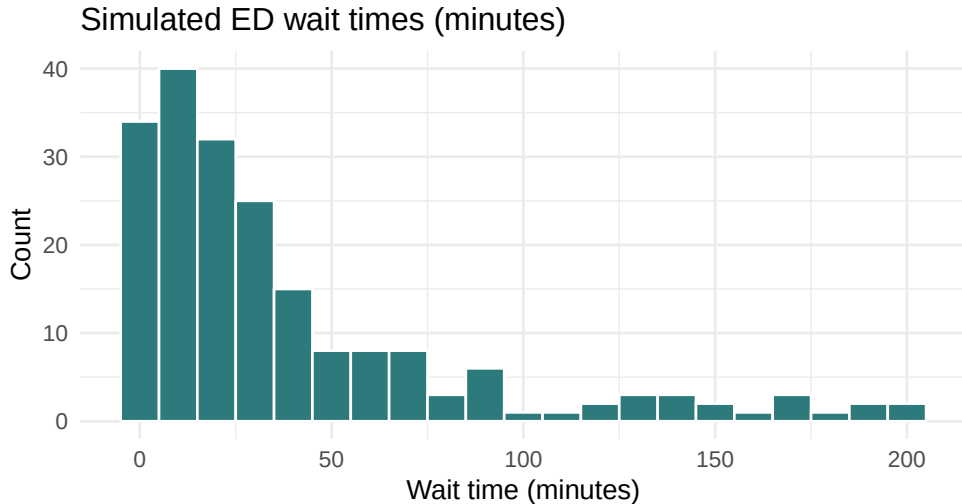
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Visual example: right skew



Long right tail $\Rightarrow$ mean pulled toward extreme delays.

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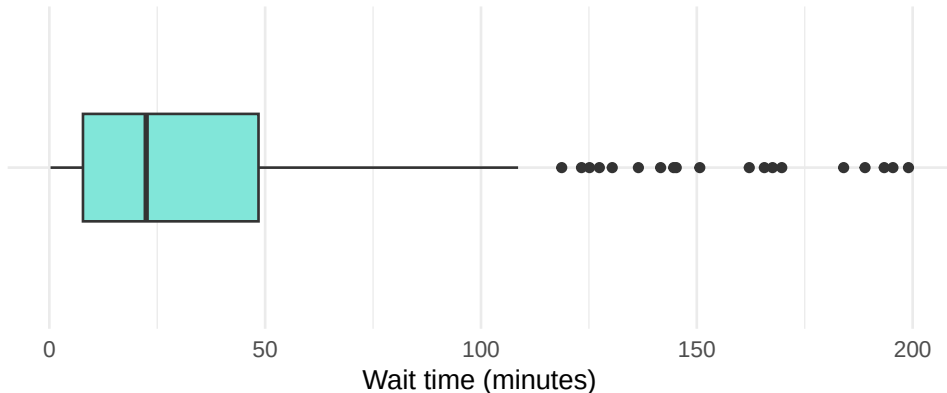
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Boxplots remind you of the five-number summary

Same waits as a boxplot



Whiskers extend to the most extreme points within about $1.5 \times$ IQR of the quartiles; points beyond are flagged as outliers.

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Scatterplots: what to say

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Describe:

- ▶ **Form:** linear / curved / no pattern
- ▶ **Direction:** positive / negative
- ▶ **Strength:** weak / moderate / strong
- ▶ **Outliers / clusters**

Name the **explanatory** (x) and **response** (y) when the scientific question implies a direction.

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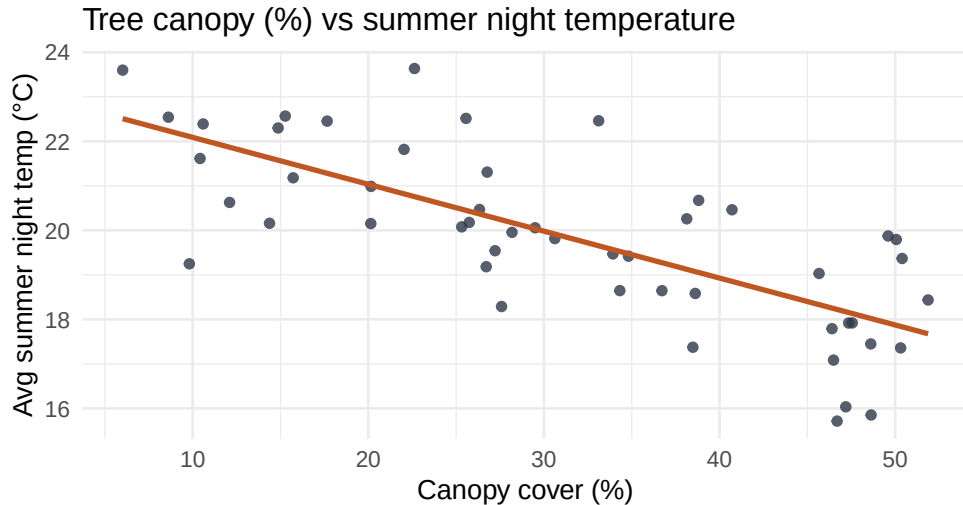
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$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

- ▶ $r \in [-1, 1]$; measures **linear** association only
- ▶ Same after linear rescaling of x or y
- ▶ Sensitive to outliers
- ▶ Association $\neq$ causation

Worked example: canopy cover and night heat

(Simulated county-style data for practice.)



Describe the relationship

Practice sentence:

*There is a moderately strong **negative linear** association between canopy cover and night temperature, with no extreme outliers.*

Compute r :

```
cor(canopy$canopy_pct, canopy$night_temp)
```

```
## [1] -0.7379817
```

r fails for nonlinear patterns

If perinatal mortality vs gestational age is U-shaped / curved, a single r can look near zero even when a strong pattern exists. **Always plot first.**

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$$\hat{y} = a + bx$$

- ▶ a : intercept (predicted y when $x = 0$) – may be meaningless if $x = 0$ is outside the data
- ▶ b : slope – average change in y associated with a 1-unit increase in x
- ▶ R^2 : fraction of variation in y explained by the linear model (for simple regression, $R^2 = r^2$)

```
fit <- lm(y ~ x, data = your_data)    # response ~ explanatory  
summary(fit)
```

Order matters: `lm(death ~ exposure, data)` not the reverse.

Fit our canopy model

```
canopy_lm <- lm(night_temp ~ canopy_pct, data = canopy)  
coef(canopy_lm)
```

```
## (Intercept)  canopy_pct  
## 23.1423450  -0.1053563
```

```
summary(canopy_lm)$r.squared
```

```
## [1] 0.544617
```

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Interpret coefficients (template)

Using whatever coefficients you got above:

- ▶ **Slope:** *For each additional 1 percentage point of canopy, predicted average night temperature changes by b °C.*
- ▶ **Intercept:** *Predicted night temperature when canopy = 0%. Check if that is near the observed range before over-interpreting.*
- ▶ **R^2 :** *About $100R^2\%$ of the variation in night temperature is explained by the linear association with canopy.*

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Prediction practice

Predict night temperature at canopy = 30%:

$$\hat{y} = a + b \cdot 30$$

```
a <- coef(canopy_lm)[1]
b <- coef(canopy_lm)[2]
a + b * 30
```

```
## (Intercept)
##      19.98166
```

or:

```
predict(canopy_lm, newdata = data.frame(canopy_pct = 30))
```

```
##           1
## 19.98166
```

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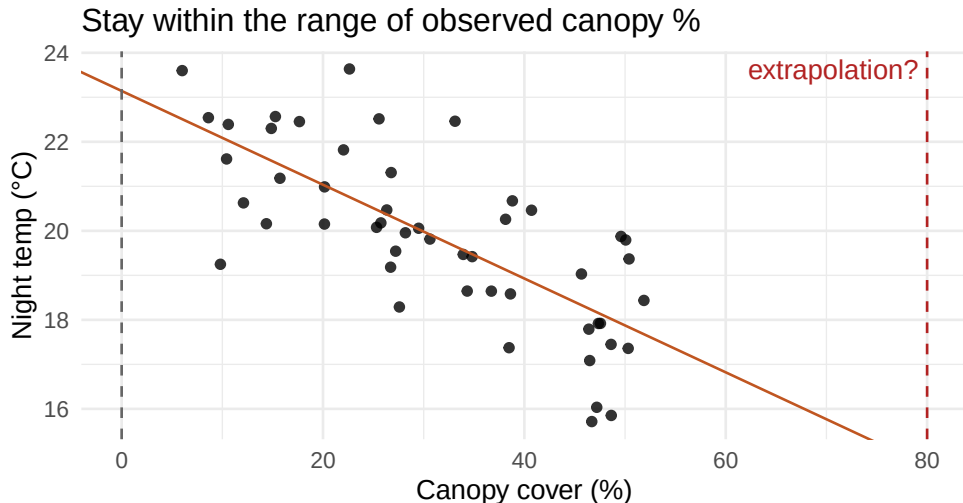
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Do not extrapolate



Predicting at canopy = 80% (or interpreting $x = 0$ if no near-zero data) is unsafe.

Fill-in-the-blank `lm()` (exam style)

Suppose we model `hr_rest` (resting heart rate) from `exercise_min` (weekly exercise minutes) in `fitness`:

```
hr_model <- _____(formula = _____ ~ _____, data = fitness)
```

Answer: `lm, hr_rest, exercise_min.`

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Legacy insurance example (optional walk-through)

If you still use the older lecture screenshot:

```
## # A tibble: 2 x 5
##   term          estimate std.error statistic  p.value
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)  10609.    1325.     8.01 2.14e-10
## 2 age           244.      33.0     7.41 1.74e- 9
```

```
insure_model <- lm(formula = charges ~ age, data = insure_subset)
```

Write $\hat{y} = a + bx$, interpret the slope, then predict charges at age 30 – same skill as the canopy example.

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A relationship between exposure and outcome is **confounded** when a third variable:

1. is associated with the **exposure**,
2. is associated with the **outcome**, and
3. is **not on the causal pathway** from exposure to outcome.

Variables on the pathway are **mediators**, not confounders.

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Coffee drinking $\leftrightarrow$ pancreatic cancer?

- ▶ Smoking may drive both coffee habits and cancer risk.
- ▶ Adjusting (or stratifying) by smoking can change the association.

Other practice pairs:

- ▶ Screen time vs GPA – sleep, course load
- ▶ Neighborhood parks vs BMI – income, transit access

Association is not causation

Even a strong r or regression slope from observational data only shows association. Causative claims need design support (randomization, careful confounding control) and subject-matter judgment.

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Two-way tables

	Disease +	Disease -
Exposed	A	B
Unexposed	C	D

- ▶ **Marginal**: overall proportions (use row/column/table totals)
- ▶ **Conditional**: proportions **within** a fixed group
e.g. $P(\text{disease} \mid \text{exposed}) = A/(A + B)$

Exams often ask you to **name** “conditional” and compute a percent.

Worked 2x2 (flu vaccination)

(Practice numbers – not an exam table.)

Group	Hospitalized	Not hospitalized
Vaccinated	18	182
Unvaccinated	42	158

Compute hospitalization risk among vaccinated and unvaccinated:

$$\frac{18}{18 + 182} = 0.09 = 9\%, \quad \frac{42}{42 + 158} = 0.21 = 21\%$$

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Why conditional?

Marginal $P(\text{hospitalized})$ mixes vaccine groups.
To compare vaccine vs no vaccine we need **conditional** risks.

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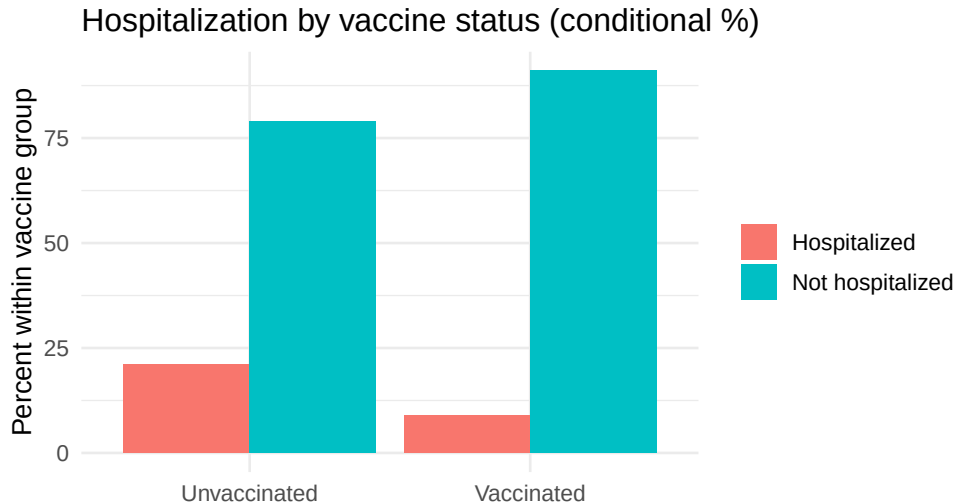
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Visualizing two categoricals

- ▶ Side-by-side (dodge) bars: compare groups easily
- ▶ Stacked bars: emphasize composition within a whole



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Simpson's paradox: communities and age

From Baldi & Moore (also used in L05): community death rates by age group.

- ▶ Within each age group, one community looks worse.
- ▶ Overall, the ranking can **reverse** because age structure differs.

Key idea: a lurking categorical variable (here age) can reverse associations when data are aggregated.

Mini Simpson table (ED discharge)

Severity	Hospital A success	Hospital B success
Mild	90/100	19/20
Severe	30/100	90/200
Overall	120/200	109/220

Check overall rates vs rates within severity – overall can favor A while each stratum favors B (or vice versa). Always stratify by the third variable when confounding is plausible.

Sampling and observational studies

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Simple linear regression

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Observational vs experimental

- ▶ **Experimental**: investigator **assigns** the exposure / treatment
- ▶ **Observational**: investigator **observes** exposure as it occurs

Confounding is especially critical for **causative** questions in observational data.

Target / study / sample

- ▶ **Target population:** whom you want to conclude about
- ▶ **Study population / sampling frame:** whom you could actually recruit from
- ▶ **Sample:** who was measured

External validity suffers when the study population is narrower than the target (e.g., only 3 universities when the target is all U.S. adults).

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How samples are drawn

- ▶ Census / complete enumeration
- ▶ Simple random sample (SRS)
- ▶ Stratified sample (proportional or oversample rare strata)
- ▶ Multistage cluster sample
- ▶ Convenience / volunteer samples – weak representativeness

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Conditioning on exposure vs outcome

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- ▶ **Cohort-style**: sample by exposure, then observe outcomes
⇒ outcome risks among exposed/unexposed are interpretable
- ▶ **Case-control**: sample by outcome (cases vs controls), then look back at exposure
⇒ compare exposure odds; cannot directly read population disease risk from the sample margins

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Randomization · Control / comparison group · Treatment

Randomization aims to balance both measured and unmeasured confounders at baseline.

It does **not** eliminate later bias from drop-out, contamination, bad measurement, or analysis choices.

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Bias after randomization (checklist)

- ▶ Under-coverage / selective recruitment
- ▶ Nonresponse; differential loss to follow-up
- ▶ Protocol non-adherence; **contamination** (controls adopt treatment)
- ▶ Measurement error; self-report bias
- ▶ Analyzing only compliers (**per-protocol**) vs **intention-to-treat**

Large n does **not** remove bias – it can make a biased estimate more precisely wrong.

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- ▶ Single / double / triple blind reduce expectancy and assessment bias
- ▶ Placebo controls help separate treatment effect from placebo effect
- ▶ Uncontrolled “everyone got the vaccine” case series lack a background rate comparison

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Be ready to spot (select-all or short answer):

- ▶ Informed consent / voluntary participation
- ▶ Coercion (especially when care access depends on enrolling)
- ▶ Equipoise (genuine uncertainty about better arm)
- ▶ Beneficence / non-maleficence / justice (Belmont)
- ▶ Data ownership, anonymization, secondary use of identifiable data
- ▶ Historical failures (e.g., Tuskegee; unethical nutrition experiments)

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A fitness-app company sells “anonymized” location trails of users to advertisers. Usernames were removed, but GPS traces still identify home and workplace.

Which classroom concerns apply?

- ▶ Data ownership / secondary use without meaningful consent
- ▶ Risk of re-identification / harm
- ▶ Beneficence / respect for persons

(Use this skill on *any* data-sharing story – do not memorize one article.)

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Researchers survey food-truck vendors for two weeks about glove use (always / sometimes / never) and record whether a health inspection later cited a hand-hygiene violation.

1. Problem type?
2. Exposure type?
3. Observational or experimental?
4. Cross-sectional or longitudinal?

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1. **Causative** – asking whether glove-use habit relates to (affects) citation risk
2. Ordinal exposure
3. Observational
4. **Longitudinal** (exposure reported first; inspection outcome later)

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Five-number summary of daily screen time (hours): 1, 2.5, 3, 4, 11; mean = 3.8.

1. Skew?
2. Better center measure if you want robustness to the 11-hour user?
3. Appropriate plot to compare screen time by year in school?

1. Right-skewed (mean $>$ median; large max)
2. Median
3. Side-by-side boxplots (or faceted histograms) – not a scatterplot of category indices, not a bar of individual continuous values

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You fit `lm(asthma_er ~ pm25, county_data)` and obtain

$$\hat{y} = 12.4 + 1.8x$$

with $R^2 = 0.36$.

1. Interpret slope in one sentence.
2. Predict ER visits per 10k at $PM_{2.5} = 8$.
3. Is predicting at $PM_{2.5} = 80$ okay if observed $PM_{2.5}$ ranges from 4-15?

Mini set 3 answers

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1. Each $+1 \mu\text{g}/\text{m}^3$ PM_{2.5} is associated with 1.8 more predicted ER visits per 10k (on average)
2. $12.4 + 1.8 \times 8 = 26.8$
3. No – that is extrapolation far beyond the data

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Before the exam, can you...

- ☐ Label descriptive / predictive / causative problems
- ☐ Name variable types (incl. ordinal vs nominal)
- ☐ Read abstracts for design labels + exposure/outcome
- ☐ Infer skew from five-number summary + mean
- ☐ Choose bar / hist / box / scatter appropriately
- ☐ Write and interpret $\hat{y} = a + bx$, predict, avoid extrapolating
- ☐ Distinguish marginal vs **conditional** distributions and compute them
- ☐ Explain confounding and recognize Simpson's paradox
- ☐ Contrast observational vs experimental; target vs study population
- ☐ Spot bias and ethics issues
- ☐ Use `select` / `filter` / `mutate` / `lm` correctly

Practice transferring skills to **new** stories.

If you only memorize one abstract or one table from class, a different cover story will feel hard – the underlying checklist stays the same.

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